

Cognitive Biology: From Systems to Somatic Networks, Tissue Cognition, Cognitive Morphogenesis, and Tissue Emotions

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Abstract

We propose a new paradigm—**Cognitive Biology**—to supersede classical systems biology. Living matter is not merely reactive machinery but an ensemble of adaptive automata that compute, represent, and pursue goals. Each cell can be understood as a *Cellutron*: a hybrid Mealy–Moore state machine whose transitions are shaped by transcriptional attention (genetic Queries/Keys/Values, after Kimmel), ion-channel admittance matrices, and signal-transduction cascades. Together, these units form *somatic networks* that exhibit emergent properties akin to cognition: decision-making, memory, and problem-solving. Morphogenesis itself is thus a form of *cognitive problem-solving in the space of shapes*, which we call *cognitive morphogenesis*. We further show that development is initialized by a *chromosomal clock*—telomere-gated PRC2 withdrawal reading a stratified, methylation-tagged genomic kernel, the hypomethylated developmental-enhancer *fossil record* of Jadhav et al. The resulting morphogenesis is formally a rectified-flow diffusion process, realized as the *expanded Neural Cellular Automaton* of Mordvintsev et al. This automaton NCA is extended with an LLM (transcriptional attention), a telomere-gated chromosomal clock, bioelectric guidance (a BETSE ControlNet after Levin), and low-rank adaptation. Its conditioning weights are *measured* super-enhancer and Activity-by-Contact maps (rather than learned through back propagation), and its denoising target is an organ atlas. We demonstrate the construction concretely: a recognizable vertebrate body plan—the conserved *phylotypic* form—computes from the inherited zygote kernel alone, with adult, species, and individual identity entering only as a low-rank adaptation layered on top. The same query–key–value feedback motif recurs across cortical attention, tissue morphogenesis, and recursive reasoning, and the deep-time elaboration of body plans appears as the evolutionary accretion of attention heads and regulatory MLPs. This framework provides a unifying theory of life-as-cognition, with implications for regenerative medicine, synthetic biology, *somatic psychology*, and artificial intelligence. Finally, extending the concept of “thermodynamic emotions for AI,” we introduce *tissue emotions*: macroscopic order parameters that summarize the energetic, informational, and control states of cognitive tissues—the closing theme this framework opens onto.

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1 Introduction

Cognitive Biology frames cells as cognitive automata (*Cellutrons*) and tissues as coupled networks that compute over developmental state spaces. This manuscript formalizes the core constructs and provides clean, orthogonally-routed figures. We proceed from the cell to the organism: Section 2 introduces the Cellutron, the chromosomal clock, and the methylation-tagged zygote kernel. Section 3 assembles these into a Large Somatic Network, exhibits its formal equivalence to a rectified-flow diffusion model, draws out the feedback–attention motif shared with cortical and reasoning systems, and tabulates the data sources that instantiate each block. This culminates in a recognizable vertebrate body plan computed from the zygote kernel alone (Figure 6, Table 1). Section 4 then reads this assembled controller back through deep time, showing how its attention heads, masks, and low-rank adapters were accreted by evolution one clade at a time—a nested LoRA

stack over the phylogenetic tree. Section 5 develops tissue emotions and somatic psychology.

2 Core Concepts

2.1 Cellutrons: Hybrid Mealy–Moore Automata

A Cellutron integrates state-dependent (Moore) and input-dependent (Mealy) behaviors under biophysical and gene-regulatory constraints. Its internal state is not static but driven by the high-velocity dynamics of the *Acetylation Engine*.

2.2 The Chromosomal Clock

We propose that the fundamental initialization protocol of differentiation is governed by a chromosomal clock (our hypotheses, 2026-01-19). This clock transduces temporal duration into spatial identity via *recursive telomere shortening*. In the zygote (x_0), telomeres are at their maximum length (approx. 10kb), promoting the expression of *TERRA RNA* which maintains the *Polycomb Repressive Complex 2 (PRC2)* in a state of high occupancy over the Hox clusters.

The deep motivation for positing such a clock is *Hox colinearity*, one of the longest-standing puzzles in developmental biology. Within a Hox cluster the genes lie along the chromosome in the same order as the territories they specify along the body: 3' genes pattern anterior structures and 5' genes posterior ones (*spatial colinearity*), and, in vertebrates, 3' genes are also switched on earlier than 5' genes (*temporal colinearity*, the “Hox clock”). Since Lewis first documented this correspondence in *Drosophila* [26], the puzzle has been why genomic order should map onto anatomical order at all—an alignment almost unique in the genome, where the chromosomal position of a gene is otherwise uncorrelated with where or when it acts [27]. The molecular signature of the process is now established: as the cluster activates, it transits gene by gene from 3' to 5' out of a Polycomb-repressed (H3K27me3) chromatin compartment and into an active one [28]. What that picture leaves open is the *timer*—what counts developmental time and drives the transition processively, in the correct direction, at the correct rate.

As the system undergoes recursion (cell division), telomeres shorten, reducing the biochemical reach of TERRA. This induces a progressive retreat of the PRC2 mask—a chromosomal scan-line—that progressively desilences Hox genes in their colinear order (3' Anterior → 5' Posterior). This temporal-to-spatial transduction establishes the body axis with algorithmic precision, ensuring that anterior identities (the *Brain Attractor*) are the first to manifest.

This furnishes the missing timer, and in doing so it resolves both faces of colinearity with a single device. Because TERRA transcription scales with telomere length and TERRA helps load PRC2 onto chromatin, the steadily shortening telomere releases PRC2 from the cluster in one monotonic 3' → 5' sweep—one increment per division. *Temporal* colinearity then follows because the sweep is clocked by division number; *spatial* colinearity follows because a processive scan-line that moves linearly along the DNA necessarily reads the cluster out in genomic order, stamping that order onto the anterior–posterior axis. The two colinearities are therefore not independent facts to be explained separately but the same monotonic process viewed in time and in space. In this account the chromosome is not merely the substrate of development but its clock: linear genomic distance along the cluster is transduced at once into elapsed time and into body position, which

is precisely why gene order and body order coincide—the long-missing mechanistic link behind Lewis’s original observation.

2.3 Epigenetic Identity: The Acetylation Engine

If the chromosomal clock writes the developmental *schedule*, a cell’s *identity* at any given moment is written in acetylation. Under our hypotheses the *Acetylation Engine*—specifically the H3K27ac mark deposited by p300/CBP and read by BRD4—is the proximate determinant of cell fate. The genome is the static *library*; acetylation is the *highlighter* that selects which passages are read now. It is a high-velocity, reversible “fluid” identity layer: unlike DNA methylation, which is copied across cell generations and serves as a long-term archive (the kernel of §2.4), H3K27ac turns over on a timescale of minutes to hours, so it encodes the cell’s *present* state rather than its developmental history. Operationally, the H3K27ac profile supplies the *Activity* term of the Activity-by-Contact score, $ABC = \text{activity (H3K27ac)} \times \text{contact (Hi-C)}$, which is precisely the attention weight from an enhancer onto its target promoter; the Promoter-MLP integrates these weighted inputs into a real-time expression profile.

Differentiation is not effected by single enhancers but by *super-enhancers*: broad, densely acetylated clusters that disproportionately control the master regulators of each cell type [24, 25]. In the attention reading of Figure 1 a super-enhancer is a pooled multi-head cluster, and the set of currently active super-enhancers *is* the cell’s identity vector. Differentiation is then a decision computed over this acetylation landscape: a progenitor scores each candidate lineage by the attention its active super-enhancers place on that lineage’s identity genes (cardiac super-enhancers $\rightarrow$ heart, neural super-enhancers $\rightarrow$ brain), and a softmax over those scores yields fate probabilities. The entropy of this distribution measures differentiation *readiness*; once it falls below a commitment threshold the cell fixes its fate, and the non-lineage super-enhancers are decommissioned—their H3K27ac removed and chromatin closed—sharpening identity at each branch of the lineage tree.

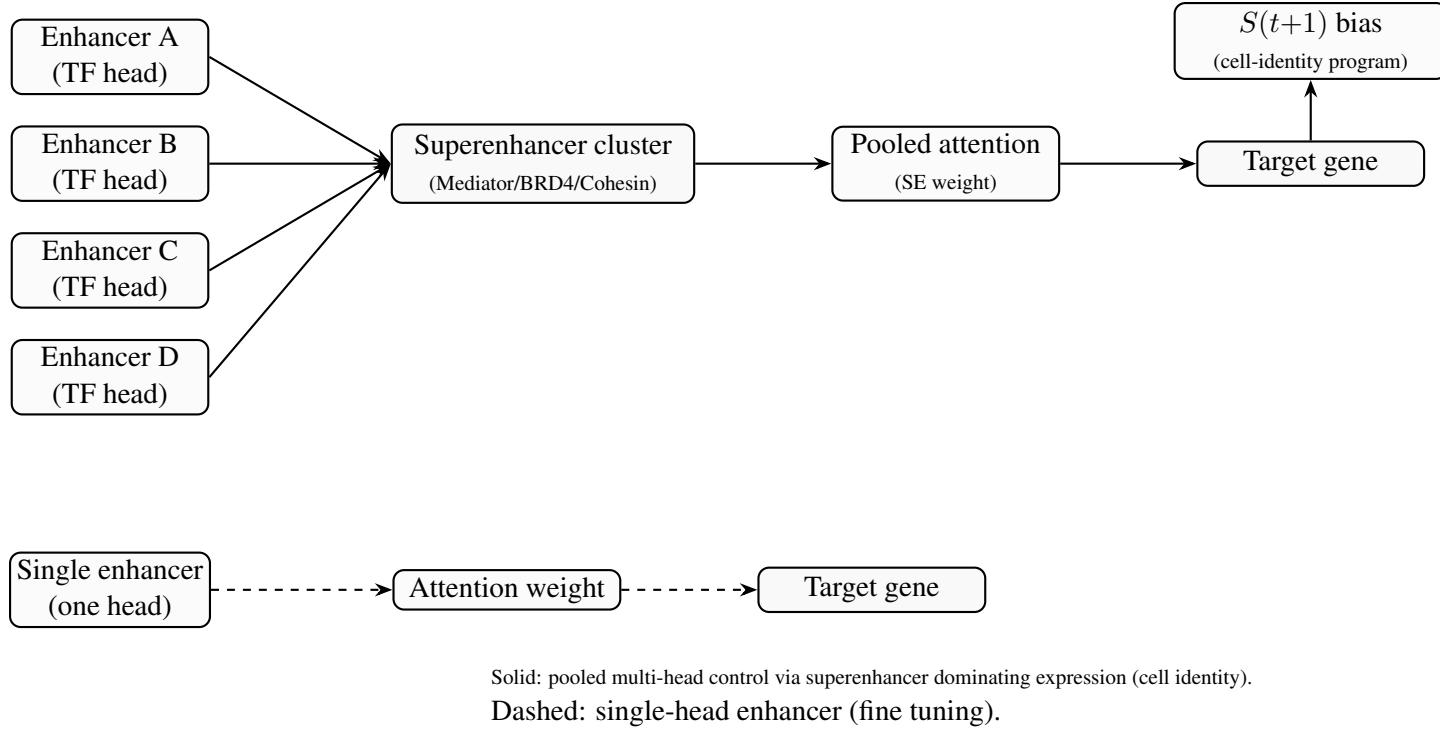


Figure 1: Superenhancers as pooled transcriptional attention heads. Multiple enhancer “heads” converge on a Mediator-rich hub, yielding a pooled attention weight that disproportionately drives target-gene expression and biases the Cellutron state $S(t+1)$.

Because acetylation is reversible, this identity can be rewritten. Pioneer transcription factors open closed chromatin and re-establish super-enhancers, and the deliberate re-acetylation of an earlier configuration is exactly what reprogramming factors (OSK/OSKM) achieve: erasing the current super-enhancer set moves the cell back up the differentiation tree, enabling the de-differentiation and re-differentiation exploited in cellular rejuvenation [29]. The *stability* of identity is therefore itself an order parameter—the breadth and persistence of the H3K27ac signal—which we later identify with the *Dominance* axis of the tissue-emotion (PAD) manifold (Section 4).

2.4 The Zygote Kernel as a Methylation Fossil Record

The chromosomal clock (§2.2) supplies *when*, and the Acetylation Engine (§2.3) supplies the active *now*; neither, however, specifies the substrate on which they operate. We identify the zygote kernel x_0 not as a flat genome but as a *stratified, tagged record* of every regulatory element the lineage will ever use—a developmental fossil record written in DNA methylation. This view is grounded in the finding of Jadhav et al. [23] that adult cells retain a near-complete molecular memory of their embryonic enhancer usage: roughly 90% of enhancers active during gut development remain *hypomethylated* in terminally differentiated adult cells long after every other chromatin feature has been erased.

The kernel therefore resolves into three superimposed strata, distinguished by which marks survive decommissioning:

- **Embryonic** (low-methylated region, LMR, only): hypomethylation is the *sole* surviving trace—H3K4me1 and H3K27ac are gone. These elements were used earliest (before E14.5) and flank developmental transcription-factor genes (e.g. the FOX family).
- **Fetal** (LMR + H3K4me1): late-acting, transient mid-gestation enhancers (E14.5–E16.5) that retain a primed monomethyl mark.
- **Adult** (LMR + H3K4me1 + H3K27ac): currently active enhancers carrying the full Acetylation-Engine signature of §2.3.

The three operations of development map cleanly onto this stratification: an enhancer is *used* (accessible, H3K4me1+/H3K27ac+), then *decommissioned* (accessibility and histone marks withdrawn, target gene silenced), but remains *tagged*—the reduced methylation is faithfully copied across cell generations and persists as memory. Of the $\sim 80,519$ such low-methylated regions catalogued in the adult, only $\sim 17\%$ are ever lost to full methylation; the remainder constitute the recoverable kernel.

Crucially, this substrate does not replace the chromosomal clock—it is *read* by it, and the two couple precisely at PRC2. In the Jadhav system, PRC2/H3K27me3 silences not the tagged enhancers themselves but the *promoters* of the bivalent transcription-factor genes; releasing PRC2—the same TERRA-gated withdrawal of §2.2—de-represses those TFs, whose products (FOX factors) then reactivate the hypomethylated enhancers *in reverse developmental order*: fetal strata first, embryonic strata second. The clock thus does not so much generate the temporal coordinate as *index into* a coordinate already inscribed in the kernel: TERRA-driven PRC2 retreat selects which methylation stratum becomes transcriptionally live, and in what sequence. This preserves the orthogonality at the heart of the framework. The methylation kernel is a *where-in-developmental-history* axis carried by the chromosome, fully independent of the bioelectric V_m map that supplies *where-in-the-body*. In the rectified-flow correspondence of Table 4, this also sharpens the identity of the seed x_0 : it is not Gaussian noise but a *structured, tagged prior*—one further respect in which the biological model departs from, and is better posed than, its image-domain counterpart.

This framing also yields a concrete rejuvenation prediction. Because the embryonic and fetal strata are decommissioned but *not erased*—they survive as hypomethylated tags—transient expression of reprogramming factors (OSK/OSKM) should reactivate them in the same PRC2-gated, reverse-developmental order described above. This is consistent with the in-vivo reversal of age-associated hallmarks by partial reprogramming [29] and with maturation-phase transient rejuvenation of human cells [30], and it casts cellular aging as drift in which strata are currently readable rather than loss of the kernel itself.

2.5 The Large Somatic Network: Multi-Scale Hypothesis

A central pillar of our hypotheses is that these processes are coordinated across a *Large Somatic Network* (or Large Somatic Model). Coupled Cellutrons form a massive, distributed intelligence that computes over morphogenetic landscapes. This network is governed by the *Tissue Recursive Model (TRM)* plus a genetic *Large Language Model (LLM)*, which uses recursive periodicity to maintain stable organ attractors and generate the growth waves observed in vertebrate development.

3 The Unified Theory of Somatic Intelligence: NCA-LLM Unification

The culmination of our hypotheses is the unification of physical morphogenesis with biological meaning through a coupled neural network architecture. We propose that biological intelligence is an ensemble of a *Neural Cellular Automaton (NCA)* and a *Large Language Model (LLM)*.

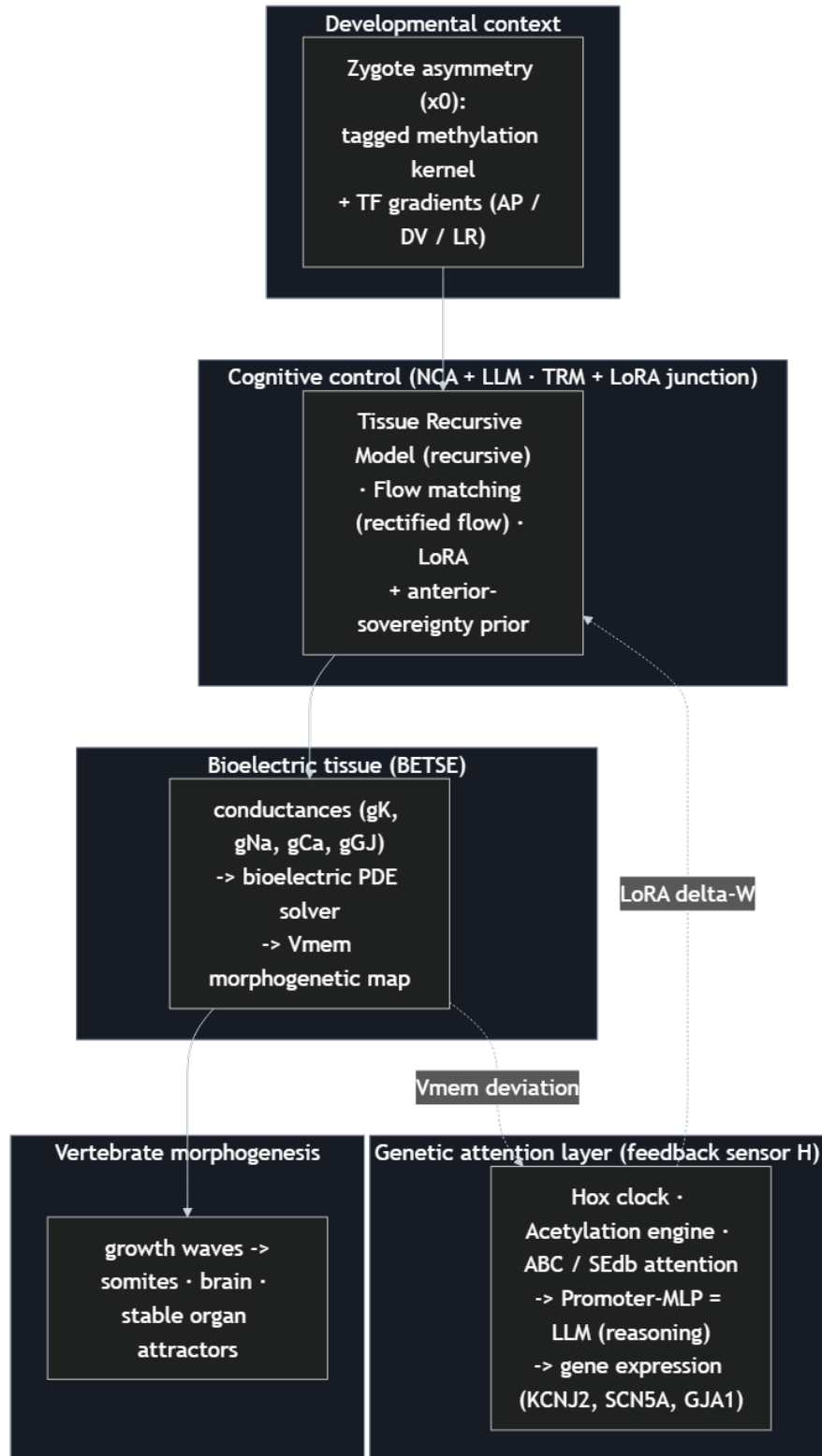


Figure 2: Overall Cognitive Biology System Architecture. The system integrates genomic timing (Telomere/Hox) with bioelectric dynamics (BETSE), governed by the Tissue Recursive Model and Flow Matching optimization to drive vertebrate morphogenesis.

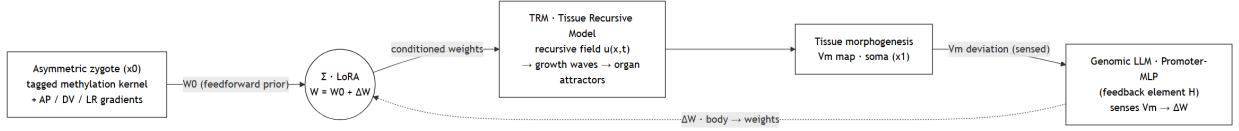


Figure 3: The architecture as a dataflow loop, in the same diagrammatic idiom as the system-architecture and transformer-style figures: the asymmetric zygote enters the LoRA summing junction (Σ) as the feedforward prior W_0 ; the Tissue Recursive Model (controller-plus-plant) maps the conditioned weights $W = W_0 + \Delta W$ to the tissue V_m morphogenesis output; and the genomic LLM (Promoter-MLP, feedback element H) senses the V_m deviation and returns the low-rank correction ΔW to the junction. Figure 4 redraws the identical loop in standard closed-loop control form.

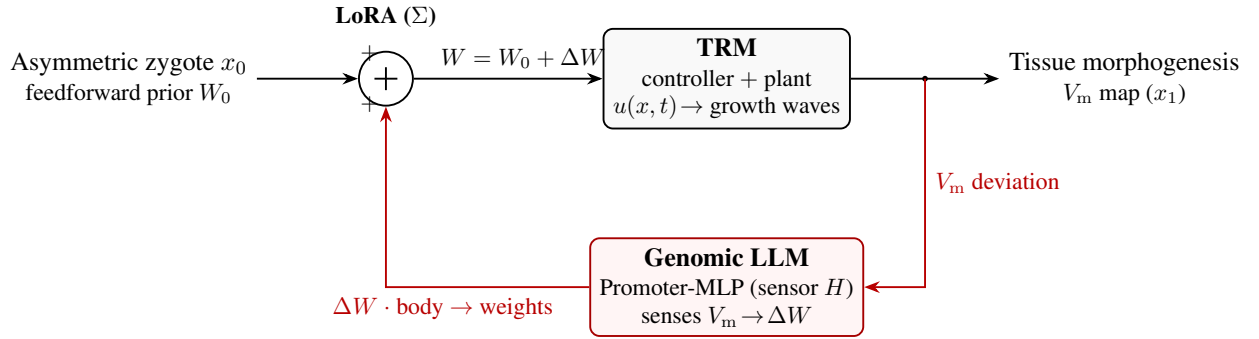


Figure 4: The same loop as Figure 3, redrawn in standard closed-loop control form: a *two-degree-of-freedom* (feedforward-plus-feedback) system. The reference r (the asymmetric zygote) enters as the feedforward prior W_0 at the summing junction Σ ; the forward path is the Tissue Recursive Model (controller-plus-plant), whose recursive vector field $u(x, t)$ integrates cell state to the output, tissue morphogenesis (V_m map, x_1). A pickoff at the output feeds the feedback element H (the genomic LLM / Promoter-MLP), which senses the V_m deviation and returns a low-rank correction ΔW along the bottom into Σ . This feedback branch (red) is the body→mind inversion of §3.4, written here as *body→weights* because the correction lands on the regulatory weights: the LoRA is the summing junction, $W = W_0 + \Delta W$, additively combining the inherited feedforward prior with the feedback correction (both +; unlike a classical error junction, the feedback adds rather than subtracts). Opening the red loop leaves the feedforward branch alone, which already computes the recognizable phylotypic form (Figure 6); closing it supplies only the species- and individual-specific remainder.

Read as a control system, the architecture is a *two-degree-of-freedom* loop (Figures 3 and 4): a feedforward branch carrying the inherited zygote kernel to the base weights W_0 , and a feedback branch—the body→mind inversion (§3.4)—in which the tissue’s V_m deviation drives a low-rank correction ΔW back onto the gene-regulatory program. The LoRA is exactly the summing junction, $W = W_0 + \Delta W$, adding the inherited prior to the live correction. This decomposition states the central result geometrically rather than decoratively: the feedforward branch alone—the loop

held open—already computes the recognizable, base-dominated phylotypic form (§3.2, Figure 6), while the feedback branch supplies only the divergent, lineage-specific remainder. The feedforward path is the conserved organism; the closed loop is its adaptation.

3.1 The NCA: The Spatial Patternner

The *Tissue Recursive Model (TRM)*—whose name and recursive deep-supervision design follow the Hierarchical Reasoning Model [31] and the Tiny Recursive Model [32]—*is* the NCA. The two are not analogous but identical, and naming both is not redundant: each name picks out a different axis of the one operator. “NCA” names its *spatial* structure—a single update rule shared across every cell (weight sharing) under strictly local neighbourhood coupling [1]; “TRM” names its *temporal* structure—that same rule applied recursively to refine the state under deep supervision, a fast inner loop within a slow outer loop [31, 32]. An NCA is already recursive in time (its update is iterated) and a TRM is already weight-shared (its tiny network is reused at every step), so the two coincide exactly when the recursion is distributed over a spatial field with local coupling—which is precisely the case here. The identity is therefore a property of *this* construction, not a definitional synonym: a generic TRM solving a non-spatial puzzle is no NCA, and a single-timescale NCA without deep supervision is no TRM. Keeping both names earns its place because each imports a result the other cannot: the NCA framing underwrites the weight-sharing identification of the kernel with the conserved genome (§3.2), while the TRM framing underwrites the recursive feedback motif by which growth waves are feedback waves (§3.4 and Tables 2–3). Its weights define the “Universal Laws of Morphogenesis.” It handles the low-level kinetic patterning, structural robustness, and growth waves. It is the distributed pattern-solver that ensures the vertebrae line up and the limbs emerge at the correct coordinates.

3.2 From Analogy to Mechanism: The Kernel Is the Conserved Genome

The defining property of a Neural Cellular Automaton is *weight sharing*: a single update rule is applied identically at every site, and all spatial differentiation arises from differences in local *state*, not in the rule. This is precisely the signature property of the genome—the same DNA resides in every cell, and identity is carried by the epigenetic state that DNA reads. We therefore make a sharper identification than analogy: the NCA kernel *is* the conserved cis-regulatory function, instantiated in our system by the conserved fossil promoter set that the hypomethylated Jadhav archive preserves—a single position-dependent map from a cell’s coordinate to ion-channel conductances and, through the Goldman equation, to a resting potential V_m . One such map is shared by the entire tissue, exactly as one update network is shared across an NCA grid.

This identification dissolves a tempting confusion. The strata of the zygote kernel (embryonic, fetal, adult; §2.4) are *not* the kernel weights—they are the clock-gated *state* the kernel reads. The TERRA–PRC2 telomere clock indexes that fossil record in reverse developmental order, deciding which strata are expressible at a given time; the conserved regulatory function that acts on them is unchanged. Weights are the genome; strata are the readable archive; promoters are the output nodes.

We make this executable in `medic/tissue/genomic_nca.py`: a tissue grid in which every cell runs the one shared genomic kernel, gated by a single `ZygoteKernel` carrying the 45,385 mark-resolved adult enhancers of Jadhav *et al.* (GEO GSE111024; 29,466 embryonic /

2,843 fetal / 13,076 adult). At full telomere length, PRC2 is fully engaged and only the adult stratum is live; the tissue expresses a heterogeneous organ voltage map (standard deviation 13.5 mV, spanning -68 to -30 mV; Figure 5). As the clock runs down, the fetal and then embryonic strata reactivate in reverse order and the voltage map collapses toward the undifferentiated zygote potential (standard deviation $\rightarrow 2.7$ mV). De-differentiation is thus reproduced as a mechanism rather than asserted: re-exposing the deepest, most conserved stratum of the methylation fossil record erases the differentiated bioelectric identity the conserved kernel had patterned.

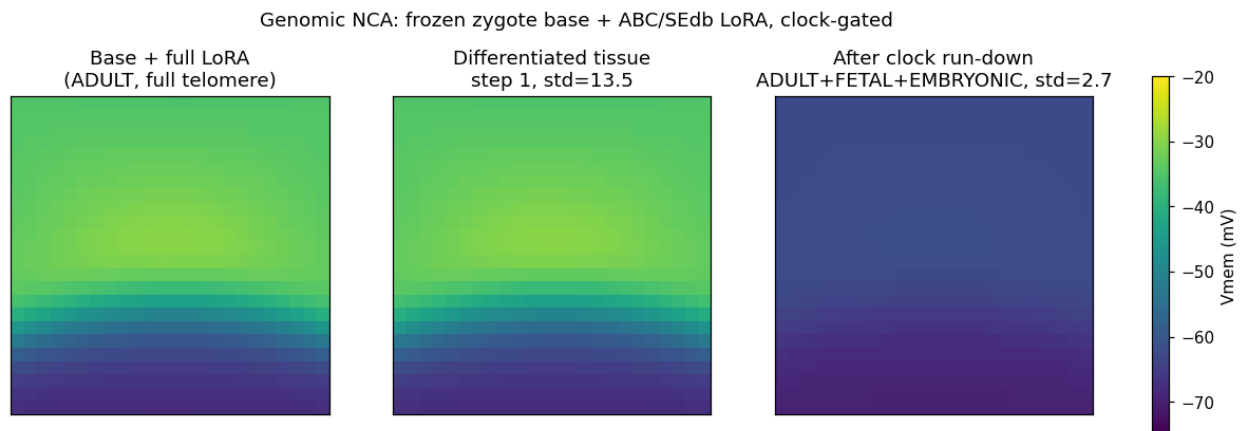


Figure 5: The NCA kernel as the conserved genome (§3.2), made executable. One conserved fossil-promoter map—the hypomethylated Jadhav archive—is shared by every cell as the conserved cis-regulatory function, while a single zygote kernel of 45,385 mark-resolved Jadhav enhancers (GEO GSE111024) is gated by the TERRA–PRC2 telomere clock. **Left:** the shared kernel’s target resting-potential map at full telomere length (adult stratum only). **Middle:** the settled, differentiated tissue (V_m heterogeneity, standard deviation 13.5 mV). **Right:** after the telomere clock runs down, the fetal then embryonic strata reactivate in reverse developmental order and the map collapses toward the undifferentiated zygote potential (standard deviation 2.7 mV)—an executable de-differentiation driven entirely by re-exposing the deepest stratum of the methylation fossil record.

Running this construction forward—from the undifferentiated zygote potential toward the differentiated map—lets us ask whether the *recognizable* form computes from the kernel. A fetus is recognizable as a vertebrate, indeed as a particular species, well before it is adult; the developmental hourglass locates that recognizability at a conserved phylotypic waist from which lineages diverge. In the LoRA coupling of §3.4 the adapter scale is precisely this hourglass coordinate: low at the waist, where the form is dominated by the inherited base, and high toward the divergent adult. We test the claim in `medic/tissue/phylotypic_form.py` (Figure 6), which assembles a body plan whose every conserved axis is *computed, not prescribed*: the dorsoventral germ-layer organization is read from the genomic kernel’s voltage field (a monotonic dorsal-hyperpolarized to ventral-depolarized gradient spanning 37 mV, recovering the Levin germ-layer polarity), antero-posterior segmentation from the *her1* clock and regressing wavefront, and limb-girdle levels from Hox chromosomal colinearity. From the kernel’s voltage prepatterning alone the assembled layout

already satisfies six of seven vertebrate Bauplan criteria—anteroposterior polarity, bilateral anterior eyes, two limb girdles in fore-anterior order, metameric trunk somites, and the anteroposterior and dorsoventral ordering of organs—and the score is unchanged when the species-specific organ-voltage residual is removed, confirming that the recognizable form is *base-dominated*: it computes from the conserved kernel rather than from the adult adapter. The one criterion the voltage cannot supply is itself informative: the bioelectric prepatter cleanly resolves the ectoderm/neural boundary (a +32 mV dorsal hyperpolarization, the most ancient and conserved bioelectric decision [36]) but leaves mesoderm and endoderm nearly isopotential (< 0.1 mV apart)—resting potential alone separates neural from non-neural. Supplying the second conserved axis, the Nodal/Activin gradient (high at the ventral pole), resolves the mesoderm–endoderm subdivision and completes the body plan at *seven of seven*. The same analysis dictated a placement correction, now implemented: the conserved dorsoventral prepatter, which initially resided in the adapter because the frozen base was a featureless uniform potential, is moved into the base W_0 (the `conserved_base` mode; §4.2), so that de-differentiation relaxes not to a structureless blank but to the conserved *phylotypic ground state*, leaving only the species-specific residual as the adapter.

Table 1: The seven vertebrate Bauplan criteria satisfied by the recognizable form (Figure 6). Each is met by a *computed*, conserved source—not a prescribed coordinate—and all seven hold on the conserved kernel alone (the species-specific residual is not required).

Bauplan criterion	Conserved computed source
Anteroposterior polarity (head before tail)	Wnt/Hox anteroposterior gradient
Bilateral anterior eyes	anterior neural (low-Wnt) eye field
Two limb girdles, fore anterior of hind	Hox colinearity ($3' \rightarrow 5' = \text{ant.} \rightarrow \text{post.}$)
Metameric trunk somites	<i>her1</i> clock + regressing wavefront
Brain anterior of gut (AP organ order)	anteroposterior axis
Neural dorsal of gut (DV organ order)	kernel V_m gradient (Levin polarity)
All organs in their correct germ layer	kernel V_m (ectoderm) + Nodal (meso/endo)

Recognizable vertebrate body plan computed from the zygote kernel (7/7 Bauplan)

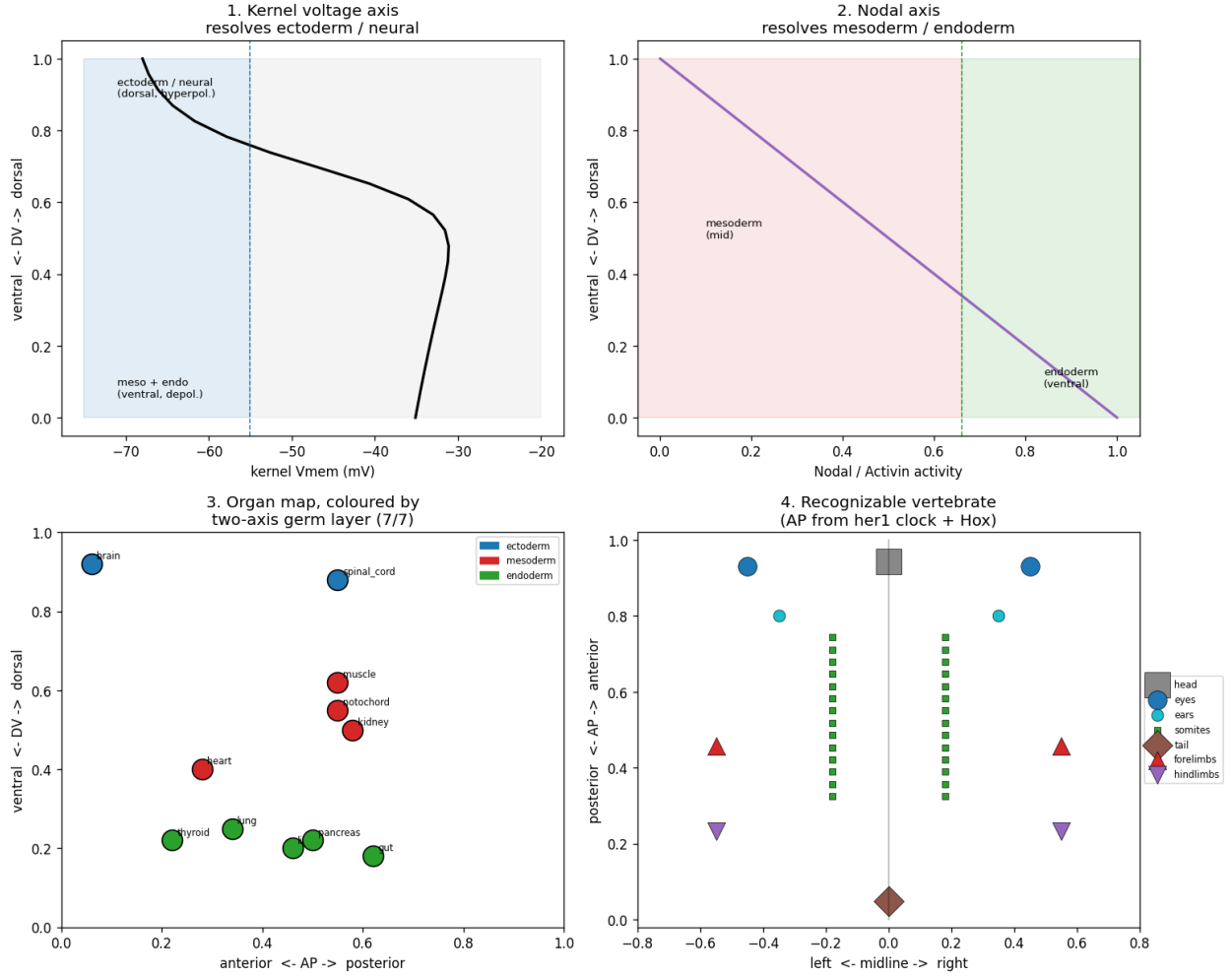


Figure 6: The recognizable vertebrate form computed from the zygote kernel (§3.2), in four panels. **(1)** The kernel voltage axis: the genomic kernel’s dorsoventral V_m profile (dorsal-hyperpolarized to ventral-depolarized, 37 mV) resolves the ectoderm/neural boundary—the most ancient bioelectric decision. **(2)** The Nodal/Activin axis (high ventral) supplies the second conserved signal, resolving mesoderm from endoderm. **(3)** The organ map, coloured by the resulting two-axis germ-layer assignment (ectoderm blue, mesoderm red, endoderm green): every organ falls in its correct layer. **(4)** The assembled recognizable vertebrate—anteroposterior somites from the *her1* clock and regressing wavefront, limb-girdle levels from Hox colinearity. Every conserved axis is computed, not prescribed; the layout satisfies all seven Bauplan criteria of Table 1 and is *base-dominated*—unchanged when the species-specific organ-voltage residual is removed—so the recognizable phyloglotypic form computes from the conserved kernel, not the adult adapter.

3.3 The LLM: The Semantic Reasoner

The *Promoter-MLP / Somatic Transformer* acts as the biological LLM (Promoter-MLP = LLM): it is the genomic language model that serves as the per-cell update rule of the patterner. It treats the genome not as a blueprint, but as a vast *pre-trained latent space*. Using Jacob Kimmel’s attention framework [22], it predicts the next state of gene expression based on the epigenetic “tokens” (Acetylation marks) and local bioelectric signals. In this context, transcription factors are Queries, the genome sequences they bind are Keys, and genes are Values; superenhancers are pooled multi-head attention clusters.

Crucially, this LLM supplies a specific *level* of biological identity. Because its substrate—the super-enhancer and Activity-by-Contact maps—is measured in adult, differentiated cells, it encodes *species- and genus-level* identity: the conserved adult regulatory wiring shared within a lineage, which the LoRA coupling of §3.4 layers onto the inherited zygote base. Genus-level identity is the part of that wiring conserved across sister species; species identity is the lineage-specific remainder. Finer, *individual* identity is not supplied here—it rides on the inherited kernel (a particular genome and methylome, including the chromosomal sex determinant) and would require conditioning on personal sequence. This division of labour is precisely why the recognizable, base-dominated phylotypic form of §3.2 computes from the kernel alone: the species- and genus-level adapter sharpens it toward a particular adult, while the recognizable organism is already present in the conserved base.

3.4 The Coupling: LoRA Adaptation

These two models are coupled via *Low-Rank Adaptation (LoRA)* [17]. The NCA (the Patterner) observes the global state and, through recursive feedback, calculates the *delta weights* ($\Delta W = B \times A$) needed to fine-tune the local LLM logic of the cells. If the physical structure deviates from the biological goal, the TRM surgically updates the weights of the Promoter-MLP to correct the course. This is a form of *Recursive Deep Supervision*, where the tissue literally teaches the cell how to regulate itself to achieve the collective shape.

That the species-specific adapter is genuinely *low-rank* is not an assumption but a measurement. Factorizing the Activity-by-Contact differentiation field over the tissue by singular value decomposition—the optimal $\Delta W = BA$ fit—a single rank-one outer product already reproduces 99.8% of its variance (root-mean-square error 1.4 mV over the resting-potential map), and rank two is exact to 0.1 mV. The adult correction the conserved kernel requires is therefore the lowest-rank update possible: one outer product layered on the inherited base. This is what makes the coupling biologically economical—the divergence from the phylotypic ground state to a particular adult is a near-trivial residual on top of a richly structured conserved kernel.

3.5 One Feedback–Attention Motif Across Brain, Tissue, and Reasoning

The attention framing is not confined to gene regulation. The same query–key–value motif, closed by a feedback loop, recurs at three scales (Tables 2–3). In the brain, the canonical cortical–subcortical circuit implements transformer attention directly: prefrontal cortex poses the query, posterior cortex supplies the keys, the association cortices carry the values, the thalamus performs the softmax routing, the cerebellum applies temporal weighting, and the basal ganglia provide

the mask (Table 2). What makes this a *reasoning* system rather than a feedforward classifier is a two-timescale feedback wave: a fast inner loop refines the current estimate while a slow outer loop supervises it (Table 3). The identical structure governs morphogenesis—fast TF–enhancer binding under slow epigenetic supervision, closed through the bioelectric field—and the Hierarchical Reasoning Model / Tissue Recursive Model [31, 32], whose low- and high-level recursions (f_L , f_H) are the same fast/slow pair under deep supervision. The central identity is therefore compact: *growth waves are feedback waves*, and developmental computation, cortical attention, and recursive reasoning are one motif instantiated in three substrates.

Table 2: Brain regions as a distributed attention head (after the “Attention Feedback Brain” mapping). The canonical cortical–subcortical loop implements query–key–value attention with thalamic softmax routing and basal-ganglia masking.

Brain region	Transformer role	Function
Prefrontal cortex	Query (Q)	cognitive intent, goals
Posterior cortex	Key (K)	sensory representations
Association cortices	Value (V)	learned features
Thalamus	Softmax / gating	routing of signals
Cerebellum	Temporal weighting	sequence alignment
Basal ganglia	Masking	action selection / inhibition

Table 3: Feedback waves and recursion across three substrates. A fast inner loop refined by a slow supervisory loop is the shared motif; in each case “growth waves are feedback waves.”

Domain	Fast loop	Slow loop	Feedback channel
Neural	gamma (30–80 Hz)	theta/beta (4–20 Hz)	cortico–thalamic
Tissue	TF–enhancer binding	epigenetic marks	bioelectric field
HRM / TRM	f_L recursion	f_H supervision	deep supervision

3.6 The Predictive Objective: A Joint-Embedding Architecture in Cell-Space

The feedback–attention motif of the previous section specifies a *mechanism*—routing and recursion—but not an *objective*. What is the recursive loop computing toward? Recent work on joint-embedding predictive architectures (JEPA) supplies a precise answer and, in doing so, sharpens what our construction already does. The proposal of LeCun and colleagues is that a world model should not predict the high-dimensional observable; it should predict in a learned latent space [2, 3]. This is easily confused with masked autoencoding, but the two differ at exactly the point that matters. A masked autoencoder reconstructs the missing *pixels* of a masked region [5]. A JEPA instead encodes the masked region into a representation and predicts that *representation*, computing its loss entirely in latent space [3, 4]. Predicting pixels spends capacity on detail that is intrinsically unpredictable—exact texture, grain, sensor noise. Predicting latents lets the model stay agnostic about that detail and commit only to the structure that is genuinely predictable: geometry, perspective, persistence, the regularities a physical world obeys. That selective discarding

is the source of the learned world primitives. Representation collapse, in which the predictor trivially emits a constant embedding, is prevented by a stop-gradient target encoder maintained as a moving average of the context encoder, together with variance regularization [4].

Read in these terms, the architecture of this paper is *itself* a joint-embedding predictive architecture, instantiated in cell-space. The “pixels” of biology are the full molecular state of a tissue—every transcript and every protein copy number—a high-dimensional observable most of whose detail is noise. Our model never predicts that observable. It predicts the next state of the bioelectric latent, the transmembrane-voltage field V_m , and both its target and its loss are defined on the organ-atlas voltage map rather than on molecular counts (§3.2). The V_m field is therefore the joint-embedding space of the construction: a low-dimensional, physically meaningful representation in which morphogenesis is predictable, while the unpredictable molecular detail is left unmodelled by design. The correspondence is exact rather than loose—the same stance that motivates I-JEPA, transposed from image patches to cell collectives.

This reading also fixes the role of the two coupled models. The outer, slow loop is a language model over the genome, which is literal sequence: an alphabet of four letters, A, T, G, and C, read by a sequence model such as AlphaGenome [18]. Its function in the architecture is that of a JEPA *encoder*—it maps genomic sequence to a regulatory latent, sequence to function—and it does *not* autoregressively generate the next nucleotide. The distinction is decisive. An autoregressive generator over bases would be exactly the construction LeCun’s argument rejects; a sequence-to-latent encoder is not. The inner, fast loop is the Neural Cellular Automaton (§3.2), the local, weight-shared spatial patternner whose efficacy for morphogenesis was established by Mordvintsev and colleagues [1]. The brain note of the previous section is then one instance of this same predictive loop, not a separate claim: cortical attention supplies the routing, predictive coding supplies the objective, and the two-timescale feedback wave—fast bottom-up error under slow top-down prediction—is the latent predictive loop that both the cortex and the developing tissue run.

3.7 Equivalence to Stable Diffusion: Rectified Flow in Cell-Space

The architecture developed above admits a direct correspondence with the modern Stable Diffusion family of generative models—specifically the Rectified Flow variant that underpins Stable Diffusion 3 and Flux [8, 9, 13]. Our Flow Matching Kernel is not analogous to that algorithm; it *is* that algorithm, applied to a different latent space. Each architectural block has a precise image-domain analogue, summarized in Table 4 and visualized side-by-side in Figure 7.

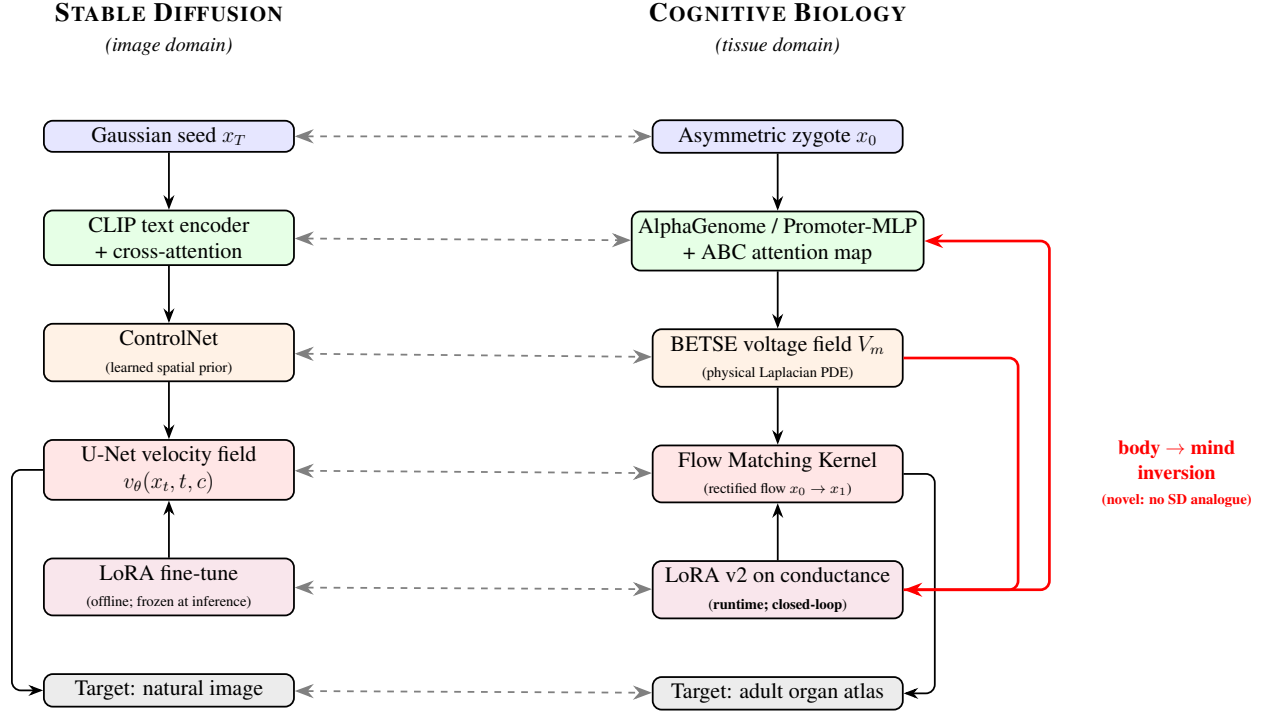


Figure 7: Side-by-side architectural comparison of Stable Diffusion (image domain) and Cognitive Biology (tissue domain). Dashed grey arrows mark the row-by-row equivalence established in Table 4; solid black arrows trace the pipeline flow within each model. The red feedback path on the right is the *body $\rightarrow$ mind inversion*: the bioelectric voltage error closes a runtime loop through the conductance LoRA back onto the gene-regulatory program embodied in the Promoter-MLP. The left column carries no analogous loop—in vanilla Stable Diffusion, LoRA is trained offline and frozen at sampling time, and classifier guidance (the closest cousin) only biases the velocity field at inference, never editing the conditioning encoder itself.

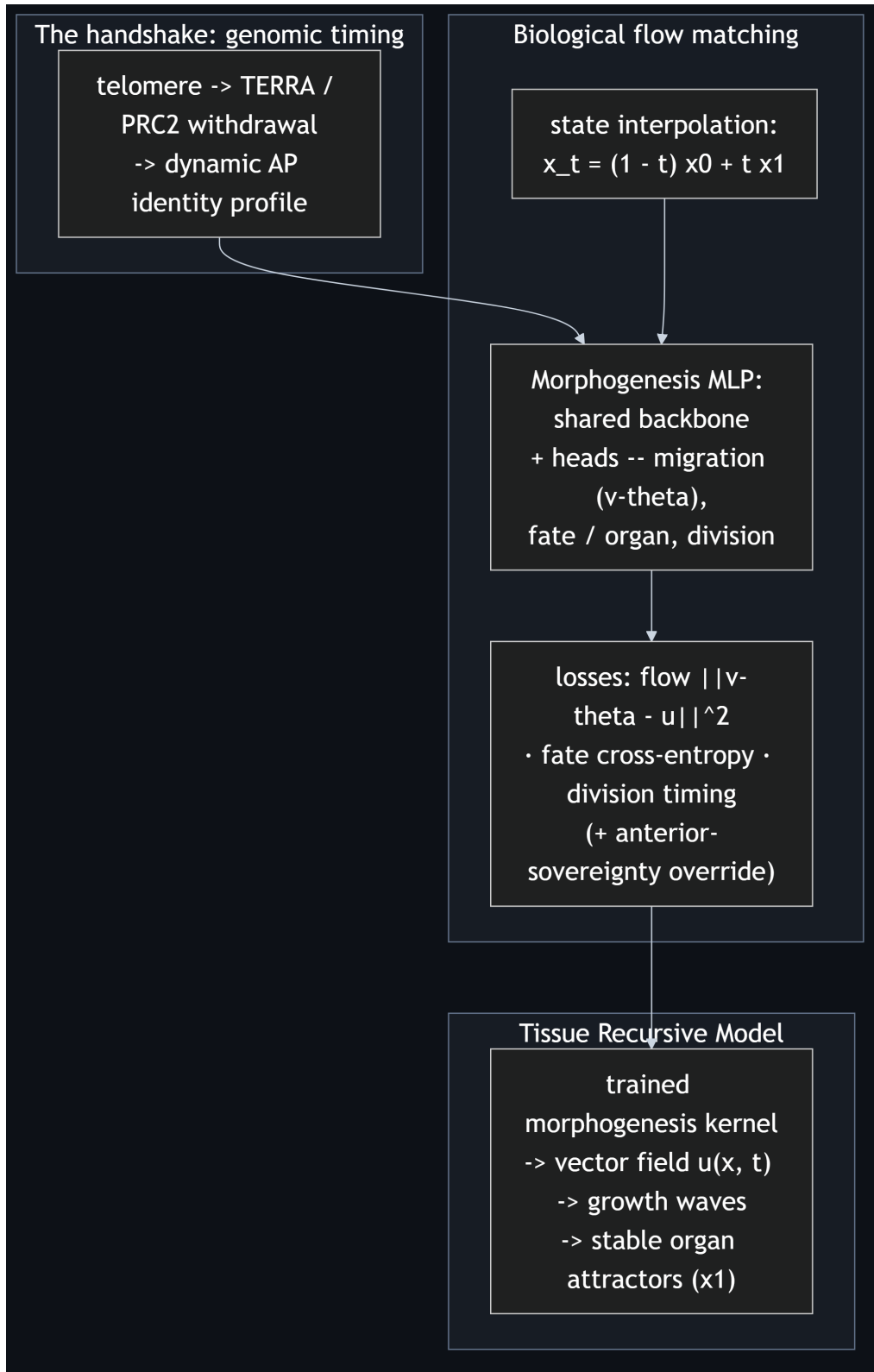


Figure 8: Biological Flow Matching Architecture. The “Handshake” (Chromosomal Clock) drives dynamic AP identity, which informs a tripartite MLP head architecture. The resulting kernel executes the Tissue Recursive Model through recursive growth waves to reach stable organ attractors.

Table 4: Architectural correspondence between Stable Diffusion (image domain) and Cognitive Biology (tissue domain). Every block of the Stable Diffusion pipeline has a biological counterpart in the proposed framework.

Stable Diffusion (images)	Cognitive Biology (tissue)
Gaussian seed x_T	Asymmetric zygote x_0 (single cell)
Pixel / latent state x_t	Cell-state vector (position, fate logits, V_m)
U-Net velocity field $v_\theta(x_t, t, c)$	Flow Matching Kernel velocity field
CLIP text embedding (conditioning c)	Gene expression profile from Promoter-MLP
Cross-attention to text tokens	ABC enhancer→promoter attention map
ControlNet (spatial conditioning)	BETSE voltage pattern V_m (Laplacian PDE)
LoRA fine-tune of U-Net	LoRA v2 on conductance layer
T denoising steps	Developmental ticks (mitosis + state update)
Target distribution: natural images	Target distribution: adult organ atlas

The biological domain provides physical anchors that the image-domain instantiations of these blocks lack. Where Stable Diffusion’s conditioning vector is an opaque CLIP embedding, ours is an observable gene-expression profile. Where its cross-attention weights are learned from text–image pairs, ours—the Activity-by-Contact map—are *measured chromatin contacts* drawn from the SEdb 2.0 / ABC database. Where its spatial conditioning module (ControlNet [14]) is another trained network, ours is the BETSE bioelectric solver: a Poisson–Nernst–Planck partial differential equation whose dynamics follow from physics rather than from data fitting. The denoising target, finally, is not a natural image but an organ atlas with known anatomy; failure modes are anatomical, not aesthetic, and the model is therefore falsifiable in a way that image-domain diffusion is not.

Concretely, the validation loop closes by rendering the simulated cell state into a synthetic histology image—Stable Diffusion 1.5 conditioned on per-region H&E prompts—and scoring that image against the reference atlas with *BiomedCLIP* [19], a domain-adapted vision–language encoder pretrained on fifteen million biomedical image–text pairs. This is the asymmetry that makes the discriminator side of the loop legitimate: CLIP plays the conditioning role *inside* the model, while BiomedCLIP plays the discriminator role *outside* it, with a vocabulary tuned to histological rather than natural-image features. Where the human target is the adult organ atlas, model organisms admit an even stronger constraint: the twelve-stage spatiotemporal atlas of zebrafish development assembled by Silic et al. [20] provides per-stage tissue maps from the one-cell zygote through prim-25, and supplies ground truth not only for the endpoint but for every intermediate state along the rectified-flow trajectory. The simulator’s checkpoints can therefore be matched against named developmental stages rather than against a single terminal distribution.

We instantiate this continuous validation concretely in `medic/zebrafish_embryo.py`: a single field of cells is carried from the zygote to the larva, so that morphogenesis and organogenesis are one simulation rather than two stitched halves. Epiboly, dorsal convergence, and convergent extension move every cell; the same cells then differentiate *in place*—somite cells settle into the chevron blocks laid down by the *her1* clock and its regressing wavefront, organ cells migrate to their Silic-onset positions coloured by the resting potential their genomic kernel specifies, and the heart field begins to beat. Figure 9 shows six stages of this one continuous field, each carrying the Silic validation score for its nearest atlas stage; coverage reaches the full mapped set and per-stage

accuracy holds in-window at 97.4% across the twelve atlas stages. Critically, the cell *number* is not prescribed but emergent (Figure 10): exact exponential cleavage to the mid-blastula transition gives way to a non-saturating power law $N(t) = 132 t^{1.91}$ ($R^2 = 0.976$ in log-log), fitted to the published Kimmel staging counts [21] and reaching $\sim 2.1 \times 10^5$ cells by 48 hpf against a real count of $\sim 2.5 \times 10^5$. A saturating Gompertz law fits the same window comparably ($R^2 = 0.973$) but is rejected on principle: it plateaus at $\sim 6.2 \times 10^4$ cells, far below the true count, whereas a real embryo continues to proliferate toward $\sim 10^6$ well past the modelled window. The proliferation schedule must therefore be modelled as open-ended—one further respect in which the developmental target constrains the model more tightly than an image-domain analogue would.

One cell field from the zygote to the larva, validated against the Silic atlas

Zebrafish embryo from the genome

3.0 hpf • cleavage / blastula (one cell field)
cells ~1,024 (drawing 6,724) somites 0



nearest Silic: 256-cell (2 hpf) ✓

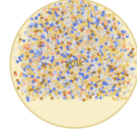
Silic validation coverage 1/2 acc 100%

one continuous cell field; pattern validated vs Silic

3 hpf - blastula

Zebrafish embryo from the genome

7.0 hpf • epiboly (one cell field)
cells ~5,417 (drawing 8,379) somites 0



nearest Silic: shield (6 hpf) ✓

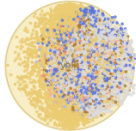
Silic validation coverage 3/5 acc 100%

one continuous cell field; pattern validated vs Silic

7 hpf - epiboly

Zebrafish embryo from the genome

10.0 hpf • gastrulation -> axis (convergent extension) (one cell field)
cells ~10,699 (drawing 9,031) somites 0



nearest Silic: bud (10 hpf) ✓

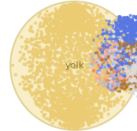
Silic validation coverage 3/4 acc 100%

one continuous cell field; pattern validated vs Silic

10 hpf - bud (100% epiboly)

Zebrafish embryo from the genome

11.0 hpf • gastrulation -> axis (convergent extension) (one cell field)
cells ~12,833 (drawing 9,208) somites 1



nearest Silic: bud (10 hpf) ✓

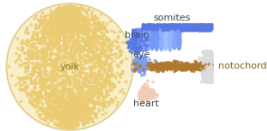
Silic validation coverage 3/4 acc 100%

one continuous cell field; pattern validated vs Silic

11 hpf - axis (convergent extension)

Zebrafish embryo from the genome

16.0 hpf • segmentation (somites + organogenesis) (one cell field)
cells ~26,232 (drawing 9,897) somites 11



nearest Silic: 12-somite (15 hpf) ✓

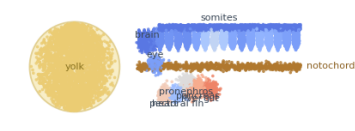
Silic validation coverage 6/6 acc 100%

one continuous cell field; pattern validated vs Silic

16 hpf - segmentation (somites)

Zebrafish embryo from the genome

48.0 hpf • pharyngula / larva (one cell field)
cells ~213,418 (drawing 12,000) somites 18



nearest Silic: long-pec (48 hpf) ✓

Silic validation coverage 9/9 acc 100%

one continuous cell field; pattern validated vs Silic

48 hpf - larva (beating heart)

Figure 9: The genome-to-embryo film as a printable montage (medic/zebrafish_embryo.py). One field of cells is carried from the zygote to the larva—blastula, epiboly, axis formation by convergent extension, somite segmentation, pharyngula organogenesis, and the beating-heart larva—so that morphogenesis and organogenesis are a single simulation rather than two stitched halves. Somite cells settle into the chevron blocks defined by the *her1* clock and regressing wavefront; organ cells are coloured by the resting potential their genomic kernel specifies (§3.2). Each panel carries the live Silic-atlas validation score for its nearest stage (coverage and accuracy), so the continuous film is scored against the twelve-stage atlas [20] at every checkpoint, not only at the endpoint.

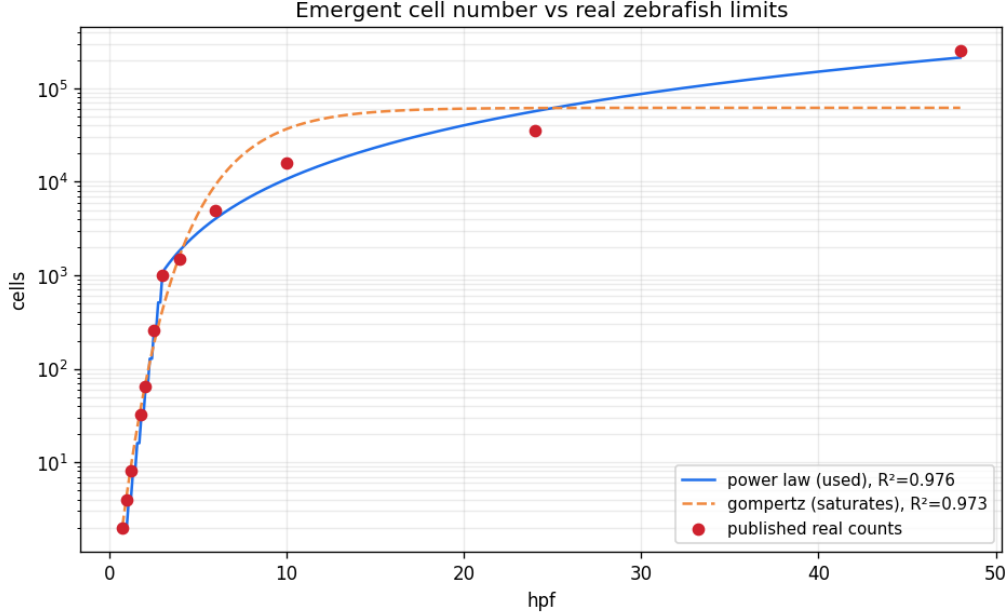


Figure 10: Emergent cell number versus real zebrafish limits. The drawn cell field grows by exact exponential cleavage to the mid-blastula transition and thereafter by a *non-saturating* power law $N(t) = 132 t^{1.91}$ ($R^2 = 0.976$; solid), fitted to published staging counts [21] (points) and reaching $\sim 2.1 \times 10^5$ cells by 48 hpf. A Gompertz law (dashed) fits the same window comparably ($R^2 = 0.973$) but saturates at $\sim 6.2 \times 10^4$ cells—far below the real count—and is therefore rejected: the embryo’s proliferation is open-ended, not asymptotic, on the developmental timescale modelled here.

The NCA framing of the kernel [1, 6, 7] contributes a biological discipline that vanilla U-Nets violate, namely locality. Standard U-Net velocity fields gain expressivity through global self-attention at coarse spatial scales, but cells in an embryo do not communicate across the body axis—they couple through nearest-neighbour gap junctions and short-range morphogen gradients. The NCA literature provides the inductive bias that matches this constraint: every cell update sees only its local neighbourhood, and the Laplacian coupling within BETSE enforces the same locality at the physics layer. The unification of NCA training and diffusion training as instances of *fitting a velocity field over time* [8, 10, 11, 12] furnishes the formal bridge: the same mathematics governs both the noise-to-image trajectory and the zygote-to-adult trajectory, differing only in the choice of source distribution, target distribution, and conditioning channel.

This perspective also identifies the natural upgrade path for the upstream *Promoter-MLP Logic* block. The block’s function is to read sequence context and produce a regulatory output—precisely the task performed by sequence-language models such as AlphaGenome [18]. Substituting AlphaGenome for the hand-engineered Promoter-MLP raises the conditioning quality of the diffusion process without altering the kernel or the PDE solver: the genome becomes the text encoder, and its predicted outputs (chromatin accessibility, expression, splicing) feed the Conductance Mapping with no further architectural change.

3.8 Data Provenance: Measured Inputs and Two External Models

Table 5 maps each architectural block to the empirical source that instantiates it. Most sources are direct substrates of the forward model or its target: mouse whole-genome bisulfite sequencing supplies the tagged methylation kernel [23]; the human SEDb 2.0 super-enhancer catalogue supplies the attention heads; the ABC database supplies the Promoter-MLP conditioning; the twelve-stage Silic atlas [20] supplies the per-stage developmental ground truth that validates the *zygote kernel*’s clock-indexed trajectory (the conserved fossil strata expressed in developmental order); and *Xenopus* bioelectric measurements supply the ground-truth voltage fields for the BETSE ControlNet. Two of the sources, however, are not training substrates but *learned models occupying the two ends of the pipeline*. **AlphaGenome** [18] sits at the front: it reads raw DNA sequence and predicts the regulatory outputs (accessibility, expression, splicing) that SEDb and ABC otherwise supply from measurement, so it can *generate the attention heads and the Promoter-MLP conditioning directly from sequence*—the genome as a learned text encoder, replacing hand-curated tracks. **BiomedCLIP** [19] sits at the back: it is the external discriminator that scores rendered synthetic histology against the reference atlas, closing the validation loop on the *output* side rather than feeding the forward pass. The asymmetry is deliberate—a sequence model conditions the generator while a vision–language model, trained on different data, judges it.

Table 5: Data provenance for the Cognitive Biology architecture. Direct sources instantiate forward-model blocks or the target atlas; the two learned models (AlphaGenome, BiomedCLIP) occupy the conditioning front-end and the discriminative back-end respectively.

Component	Role	Source
Tagged DNA kernel (x_0 strata)	methylation fossil record	Mouse WGBS (Jadhav; GEO GSE111111)
kernel developmental trajectory	per-stage ground truth (validation)	Silic zebrafish (12 stages)
Attention heads (enhancers / SEs)	cross-attention map	SEdb 2.0 (human)
Promoter-MLP logic	regulatory reasoning	ABC database
Bioelectric field V_m (ControlNet)	spatial conditioning truth	Xenopus electric-field data
Sequence→function encoder	conditioning front-end	AlphaGenome (learned)
External discriminator	output validation vs. atlas	BiomedCLIP (learned)

4 Evolutionary Assembly of the Controller

Having assembled the full Large Somatic Network as an architecture—the coupled NCA + LLM pattern of the preceding section—we now read its components back through deep time. The same attention heads, masks, and low-rank adapters were not designed at once but accreted by evolution one clade at a time.

4.1 Evolutionary Accretion of Attention Heads and Regulatory Logic

If enhancers are single attention heads and super-enhancers are pooled multi-head clusters (Figure 1), then the deep-time growth of animal complexity reads as a single process: evolution did not

rewrite the attention mechanism, it *accreted heads, duplicated them, and added masks and MLPs*. Simpler organisms run strict subsets of the full controller. Table 6 traces this sequence, in which each clade contributes one architectural component.

Table 6: Phylogenetic accretion of the morphogenesis controller: each clade adds one component (head, mask, or MLP), and simpler organisms run strict subsets. Body-plan and brain complexity track the number and pooling depth of transcriptional attention heads.

Era / clade	Innovation	Component added
Prokaryotes (~ 3.8 Ga)	ion channels, V_m	Body-Plan Head (1)
Eukaryotes (~ 2.0 Ga)	histones, PRC2	Chromatin Head + Polycomb Mask
Sponges (~ 800 Ma)	Wnt signalling	Morphogen Head (2; oral–aboral)
Cnidarians (~ 600 Ma)	gap junctions, 2nd axis	Connexin Head (3)
Bilaterians / insects (~ 550 Ma)	Hox cluster, segments	Polycomb Mask becomes temporal (clock)
Vertebrates (~ 500 Ma, $2\times$ WGD)	4 Hox clusters, neural crest	full 6 heads + SE Head + all masks
Amniotes / reptiles (~ 300 Ma)	elaborate enhancers	Promoter MLP
Mammals (~ 200 Ma)	diverse ion channels	Channel MLP

The two rounds of whole-genome duplication at the base of the vertebrates—copying each Hox cluster from one to four—are, in this reading, the single largest parameter-expansion event in the history of life: the same rewrite rules instantiated with four times the attention heads, yielding neural crest, jaws, paired appendages, and complex organs. Under our hypotheses this is simply the evolutionary face of the same architecture—the chromosomal clock and the tagged kernel supply the developmental schedule and substrate, while evolution tunes *capacity* by accreting heads, masks, and MLP width. The same logic governs nervous-system complexity. The diffuse nerve net of the radially symmetric cnidarians (*Hydra*, jellyfish, sea anemones) runs only the earliest heads on a single oral–aboral axis. The centralized brains of bilaterians—insects such as *Drosophila*—require the Hox/segmentation clock that polarizes the anterior–posterior axis. And the elaborate enhancer and ion-channel landscapes of amniotes (reptiles) and then mammals supply the regulatory depth—the Promoter and Channel MLPs—that underlies cortical expansion. Morphological and neural complexity thus track the *number and pooling depth of transcriptional attention heads*, not coding-gene count.

4.2 Phylogeny as a Nested LoRA Stack

If the accretion of Table 6 adds attention heads on the semantic side, the same deep-time process has a counterpart on the substrate side: the tagged zygote kernel of §2.4 is itself *clade-specific*—a cnidarian gastrula, an arthropod germ band, and a vertebrate body plan freeze radically different developmental archives—and derived clades inherit their ancestors’ kernels and *expand* them. We formalize this as a nested composition of low-rank adapters [17]. Writing W_0 for the conserved base kernel and ΔW_c for the regulatory innovation contributed by clade c , the effective kernel of a vertebrate is

$$W_{\text{vert}} = W_0 + \Delta W_{\text{bilateria}} + \Delta W_{\text{deuterostome}} + \Delta W_{\text{chordate}} + \Delta W_{\text{vertebrate}}, \quad (1)$$

each ΔW_c a residual adapter that is the identity at the moment of branching. Because the adapters are additive and zero at their branch point, introducing a clade is a no-op for its ancestor: the construction is convergence-preserving, and an ancestral developmental program is recovered exactly by zeroing the later adapters. The branching is essential. Cnidarians, flatworms, arthropods, and vertebrates are not a linear ladder but lineages diverging from a shared bilaterian prefix, so the adapter stack itself *branches*—protostomes and deuterostomes share $W_0 + \Delta W_{\text{bilateria}}$ before diverging—and that shared prefix is exactly the conserved cis-regulatory function (§3.2). This is the molecular signature of *deep homology* [35]: the same toolkit genes (*Pax6*, the Hox cluster) building homologous structures across phyla are the shared base weights, frozen before the Cambrian.

Two floors of this stack reach back across very different timescales. The base W_0 —the ion-channel cis-function whose K^+ selectivity filter and Nernst potentials are invariant from bacteria to mammals—predates animals entirely, traceable to the bioelectric machinery of prokaryotes (~ 3.5 Ga; Table 6). Above it, the enhancer fossil strata of §2.4 begin only with metazoa (~ 800 Ma) and accrete per clade, the vertebrate whole-genome duplications (Table 6) furnishing the largest single expansion. The fossil record of *channels* thus runs far deeper than the fossil record of *enhancers*.

Two cautions keep the claim well-posed. First, the nesting is the conserved *phylotypic* core of von Baer’s laws [33] and the developmental hourglass [34], not Haeckelian recapitulation: embryos share a conserved mid-developmental waist and diverge on either side of it; they do not replay ancestral adults. Second, the physical archival tag is not universal. CpG hypomethylation as durable enhancer memory is largely a vertebrate property; many invertebrates methylate sparsely or (as in *Drosophila*) almost not at all, and likely archive developmental memory in other inherited chromatin states such as Polycomb domains. The architectural claim is therefore substrate-independent—each clade freezes a developmental archive and derived clades expand it—while the readout that instantiates a given clade’s adapter must be matched to that clade’s chromatin. In our implementation this is precisely the role of the external adapter in the LoRA coupling (§3.4): a frozen, inherited base records the developmental program while a clade-appropriate readout supplies each ΔW_c .

5 Tissue Emotions and Somatic Psychology

When we move from cells to tissues, ensembles of Cellutrons couple through bioelectric, metabolic, and mechanical interactions to form *somatic networks*. We define tissue emotions as emergent thermodynamic order parameters summarizing the energetic and control states of such a Large Somatic Network. These are formalized via the **PAD Manifold**:

- **Valence (V)**: Homeostasis vs. stress, measured by stress gene signatures and metabolic efficiency.
- **Arousal (A)**: Excitability level, measured by ion channel activity and Ca^{2+} dynamics.
- **Dominance (D)**: Identity lock-in, measured by superenhancer breadth and the stability of the H3K27ac signal.

The PAD state biases which attractor basin the system enters in the morphogenetic landscape. We model this navigation using Langevin dynamics:

$$dX = -\nabla U_\phi(X)dt + \sigma dW \quad (2)$$

Where U_ϕ is a potential function parameterized by the PAD state, and σdW represents the developmental noise. This framework provides the mathematical basis for *somatic psychology*, where maladaptive PAD states (e.g., negative valence, high arousal, low dominance) tilt the system toward tumorigenic or fibrotic attractors.

Methods

Our modeling strategy integrates multi-layer data and inference through the following pipeline, formalized as *our hypotheses*:

1. **Chromosomal Clock Calibration:** We calibrate the recursive telomere shortening rate to the developmental timeline. The withdrawal of PRC2 silencing is modeled as a colinear scan-line that activates Hox-dependent enhancers in sequence.
2. **Large Somatic Network/Model:** Central to our hypotheses is the coordination of developmental processes across a *Large Somatic Network*. Cellutrons are treated as nodes in a distributed computational fabric, where collective decision-making governs global morphology.
3. **Epigenetic Fate Determinism:** We define the *Acetylation Engine* (specifically H3K27ac) as the *only* determinant of cell fate. The dynamic acetylation profile acts as the highlighter that identifies active regulatory nodes within the ABC (Activity-by-Contact) attention database.
4. **Promoter-MLP Logic** (Promoter-MLP = LLM): Transcription factor binding affinity shifts (ΔPWM) are processed through the Promoter-MLP reasoning engine—the genomic language model. This logic computes real-time gene expression profiles (*KCNJ2*, *SCN5A*, *GJA1*) which are then mapped to ion-channel conductance parameters (g_K , g_{Na} , g_{gj}).
5. **Bioelectric Simulation:** The BETSE (Bioelectric Tissue Simulation Engine) solver integrates these conductances via a Laplacian coupling PDE to generate morphogenetic voltage patterns (V_m).
6. **Tissue Recursive Model (TRM):** Within the Large Somatic Model, the system maintains a recursive feedback loop where emerging voltage patterns are compared against stable adult organ attractors (x_1). This model generates the characteristic growth waves and rhythmic periodicity of vertebrate morphology.
7. **Flow Matching Kernel (the three heads):** To execute the transition from zygote (x_0) to adult (x_1), we employ a tripartite *TorchMorphogenesisMLP* kernel trained via Flow Matching (Rectified Flow). It is important to be clear about *where these heads are situated*: this kernel *is* the per-cell update rule of the NCA/TRM (§3.2, §3.6)—one weight-shared network, run identically at every cell, whose three output heads are exactly the three decisions a cell

makes at each recursive step. They are therefore not three separate networks but three read-outs of a single shared kernel, which is why migration, fate, and division remain mutually consistent across the whole tissue:

- **Migration Head** (v_θ)—*the where*: the rectified-flow velocity field that transports each cell along its trajectory, carrying the zygote distribution toward the adult layout. It is the head responsible for convergent extension, germ-layer sorting, and the placement of organs at their correct coordinates.
- **Fate Head**—*the what*: it scores the cell’s current acetylation-driven identity (the H3K27ac highlighter reading the ABC attention map) against the target organ identity by cross-entropy, committing the cell to a lineage as the super-enhancer attention sharpens.
- **Division Head**—*the whether*: it gates the temporal schedule of proliferation against the developmental clock, producing the open-ended power-law growth in cell number rather than a fixed lineage tree.

The *Anterior Sovereignty Override* applies a 5.0 scalar weight to the Fate Head’s loss for neural regions, ensuring the brain attractor is secured with maximum fidelity. Situated this way, the three heads are the action-output layer of the genomic language-model cell rule (Promoter-MLP = LLM): the LLM reasons over the epigenetic tokens and emits, through these heads, the cell’s movement, fate, and division—the readout of the NCA/TRM update applied at every node of the Large Somatic Network.

8. **Projection to PAD Manifold**: The resulting tissue states are mapped to Valence, Arousal, and Dominance order parameters (Tissue Emotions), which bias the stochastic Langevin trajectories toward regenerative attractors.

Relation to Prior Work

Jacob Kimmel has described an attention-like framework for gene regulation [22], in which transcription factors are queries, the genome sequences they bind are keys, and genes are values; because identity is then carried by these embedding vectors, editing a single one (a binding sequence) can produce large, targeted changes in output, which is what makes transcription factors such efficient reprogramming levers. This *in silico* view of epigene circuits—used to reproduce cell-type identities and to design synthetic regulatory interventions—aligns closely with our definition of Cellutrons and somatic networks.

A second lineage of prior work concerns the *spatial* substrate. Mordvintsev and colleagues’ *Growing Neural Cellular Automata* [1, 6, 7] established that morphogenesis can be cast as a single local update rule, shared across a field of cells, that grows and regenerates a target pattern from one seed under nearest-neighbour communication alone—morphogenesis as distributed computation. Our patterner is a direct descendant of that construction (it *is* an expanded NCA, §3.2), and we extend it in three respects. Where their rule is learned end-to-end by back-propagation against a chosen image, ours is largely *measured*: its conditioning weights are super-enhancer and Activity-by-Contact maps rather than fitted parameters. Where their seed is a blank single cell, ours is the structured, methylation-tagged zygote kernel (§2.4). And where their target is an emoji, ours is an

organ atlas with falsifiable anatomy, reached under a chromosomal clock and a physical bioelectric field that their toy automata lack.

A third lineage concerns the *spatial guide* itself. Michael Levin’s work on molecular bioelectricity [36] established that endogenous transmembrane voltage gradients (V_m) are not byproducts of physiology but *instructive, non-local pattern-control signals*: imposed voltage prepatterns can redirect regeneration and organ placement in vivo, and the resting-potential map carries a spatial code read by downstream gene expression. This bioelectric code is exactly the ControlNet of our architecture (Table 4): we formalize it as the BETSE Poisson–Nernst–Planck conditioning field, show that the most ancient bioelectric decision—the ectoderm/neural boundary—computes from the conserved kernel’s V_m prepattern (§3.2, Figure 6), and extend it by closing the loop from the voltage field back onto the gene-regulatory program through the body→mind LoRA inversion (§3.4), a coupling Levin’s descriptive framework anticipates but does not render computationally.

A fourth lineage concerns the *objective* that our feedback–attention motif (Tables 2–3) optimizes. Recent joint-embedding predictive architectures [2] argue, with LeCun, that cortical computation is better cast as prediction in a learned latent space than as generative modelling of the observable; this is compatible with rather than opposed to our mapping, since attention supplies the *routing* (how) while predictive coding supplies the *objective* (why), and our two-timescale feedback wave—fast bottom-up error under slow top-down prediction—is itself such a latent predictive loop. Our morphogenesis engine takes the same stance, predicting the next state in the bioelectric (V_m) latent rather than generating tissue observably (§3.2), so that the framework is in effect a biological joint-embedding predictive architecture of which the cortical attention loop is one instance.

However, our Cognitive Biology framework extends beyond regulatory circuits, self-organizing automata, and bioelectric pattern control taken individually; its contribution is to unify them. First, we introduce the PAD manifold (Valence, Arousal, Dominance) as emergent thermodynamic order parameters of tissues—interpreted as tissue emotions. Second, we embed these parameters into a morphogenetic landscape, allowing tissues to be modeled as cognitive systems that navigate attractors corresponding to regeneration, tumorigenesis, or fibrosis. Third, we integrate GWAS variation directly into the enhancer–TF–gene mapping, permitting in silico perturbations that predict attractor bias. Finally, our approach emphasizes tissue-level cognition and somatic psychology, whereas prior work has focused primarily on transcriptional circuit design, on self-organization in the abstract, or on bioelectric signalling in isolation.

Conclusion

Cognitive Biology, as developed in this manuscript, can be summarized in a single technical sentence: it is a cross-attention-conditioned, ControlNet-guided, LoRA-finetuned rectified-flow diffusion model whose latent space is cells, whose text encoder is the genome (read by a sequence model such as AlphaGenome [18]), whose ControlNet is the BETSE bioelectric solver, and whose denoising target is an adult organ atlas. Every architectural block has a Stable Diffusion analogue [8, 9, 13], and the unification of Neural Cellular Automata with diffusion-style velocity-field models [1, 6, 11, 10] provides the formal bridge. The biology is what makes the model falsifiable: the conditioning signal is observable, the cross-attention weights are measured rather than learned, and the spatial guide is physically simulated rather than fit to data. At the centre of that bridge is a

single operator wearing two names: the patterner is at once a Neural Cellular Automaton, named for the spatial weight-sharing it shares with the genome, and a Tissue Recursive Model, named for the temporal recursion it shares with cortical and reasoning feedback (§3.2). That the kernel *is* the conserved genome and that growth waves *are* feedback waves are therefore not two findings but one—the same operator read along its spatial and its temporal axis.

Stated plainly as an architecture, the system is an **NCA + LLM** pair. The patterner is a Neural Cellular Automaton—a single weight-shared rule run locally at every cell—and that automaton, run under recursive deep supervision (the fast inner loop refined by a slow outer loop), *is* the Tissue Recursive Model: $\text{NCA} + \text{deep supervision} = \text{TRM}$. Its per-cell update rule is not a generic multilayer perceptron but a genomic language model that reads epigenetic tokens and bioelectric context: the Promoter-MLP, where $\text{Promoter-MLP} = \text{LLM}$. The two are joined by the LoRA coupling of §3.4, the patterner computing the low-rank correction that fine-tunes the language model’s per-cell logic. In the notation of §3.2 and §3.6, Mordvintsev’s $\text{NCA}(\text{MLP})$ —a plain rule grown to a target image—becomes our $\text{TRM}(\text{LLM})$: the same local, weight-shared automaton, its cell rule upgraded to a genomic language model and its iteration supervised across two timescales, predicting in the bioelectric latent rather than generating the molecular observable. The three output heads of that shared rule—migration, fate, and division—are where the language model’s reasoning becomes a cell’s action.

One element of the architecture, however, does not reduce to any existing Stable Diffusion construction. In vanilla diffusion, gradients flow from the data loss to the U-Net weights, full stop. In Cognitive Biology, the gradient flow includes a closed loop: the tissue’s bioelectric voltage error drives a low-rank update on the conductance layer (the LoRA inversion), which then propagates back to the upstream gene-regulatory program embodied in the Promoter-MLP. This is the *body* $\rightarrow$ *mind inversion*: physiology rewriting its own gene-regulatory prior at runtime, in response to the morphological deviation the tissue is currently experiencing. The closest analogue in the generative-modeling literature is classifier guidance [15, 16], in which an external signal steers the denoiser at sampling time; but the LoRA-into-the-regulatory-prior coupling, in which the spatial dynamics modify the conditioning network’s own weights rather than merely biasing its outputs, is genuinely novel. It is the architectural feature that distinguishes Cognitive Biology from the image-domain models whose mathematics it inherits, and it is the proper subject of the next paper in this series.

Underlying all of this is the *zygote kernel*. If the rectified-flow machinery is the engine, the kernel is what the engine reads: the inherited, methylation-tagged *fossil record* of every developmental enhancer a lineage has ever used (§2.4), frozen as hypomethylated DNA and indexed in reverse developmental order by the telomere–PRC2 clock. It is, in this framework, the most conserved and most defining object in the architecture—the substrate from which the recognizable organism is computed *before* any adult, species, or individual detail is layered on. We have shown this concretely: a recognizable vertebrate body plan emerges from the kernel alone (§3.2, Figure 6), the super-enhancer and Activity-by-Contact adapters supplying only the divergent, lineage-specific remainder as a low-rank correction. Across phyla the kernel is itself a nested stack of such corrections (§4.2), so that to read the kernel is to read deep time. The conserved, recognizable organism—the phylotypic self—is written in the kernel; everything else is adaptation. Whatever else Cognitive Biology becomes, the zygote kernel is its conserved core.

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